Math 55 worksheet [Monday/Wednesday] [MM]/[DD]/2026

1. How many non-negative integer solutions are there to the equation $a + b + c + d = 15$?
   (a) In general.
   (b) If $a \geq 4$.
   (c) If $a \geq 4$ and $c \leq 8$.

2. How many ways are there to fill a $2 \times n$ box with n dominoes of size 1×2, if each domino can be placed either vertically or horizontally and no dominoes are allowed to overlap?

3. Using the binomial theorem, what is the coefficient of x^6y^{10}
   (a) in $(x + y)^{16}$?
   (b) in $(4x + 3y)^{16}$?
   (c) in $(4x + 3y^2)^{11}$?

4. Give a combinatorial proof of Vandermonde's identity:

$$\binom{m+n}{r} = \sum_{i=0}^{r} \binom{m}{i} \binom{n}{r-i}$$